

DEFINITIONS

FROM AN EM POV, ANY POSITION IN SPACE AND TIME IS CHARACTERIZED BY THREE TIME VARYING SCALAR FIELDS:

- DIELECTRIC PERMITTIVITY $\epsilon(\underline{r}, t)$ $[\frac{F}{m}]$
 - MAGNETIC PERMEABILITY $\mu(\underline{r}, t)$ $[\frac{H}{m}]$
 - CONDUCTIVITY $\sigma(\underline{r}, t)$ $[\frac{S}{m}]$
- THESE PARAMETERS WOULD BE DIFFERENT BASED ON MEANS OF INTERPRETATION.
E.G. ACOUSTIC WAVES (C: STIFFNESS, ρ : DENSITY)

REFERENCE SCENARIO

FROM THE EM POV, THE REFERENCE SCENARIO IS MADE OF SOME DISTRIBUTION OF $\epsilon(\underline{r}, t)$, $\mu(\underline{r}, t)$, $\sigma(\underline{r}, t)$ INSIDE D_{inv} .
SINCE IT IS A REFERENCE, THIS DISTRIBUTION MUST BE KNOWN!

IN MANY CASES THE REFERENCE SCENARIO IS ASSUMED TO BE HOMOGENEOUS

$$\epsilon(\underline{r}, t) = \epsilon \quad \mu(\underline{r}, t) = \mu \quad \sigma(\underline{r}, t) = \sigma \quad \forall \underline{r} \in D_{inv}$$

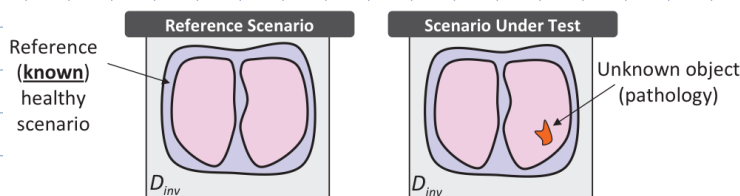
IN THE PARTICULAR CASE OF VACUUM, THESE CONSTANTS ARE KNOWN

$$\epsilon(\underline{r}, t) = \epsilon_0 \quad \mu(\underline{r}, t) = \mu_0 \quad \sigma(\underline{r}, t) = \phi \quad \forall \underline{r} \in D_{inv}$$

FROM NOW ON, WE'LL CONSIDER A HOMOGENEOUS REFERENCE SCENARIO, ALSO CALLED FREE-SPACE.

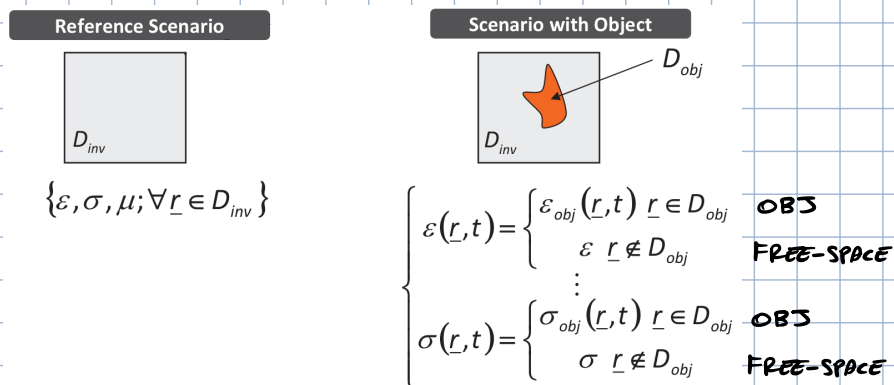
EXAMPLE: BIOMEDICAL

WE MUST KNOW WHAT HEALTHY LUNGS LOOK LIKE TO FIND ISSUES $\rightarrow$ EXPERIENCE

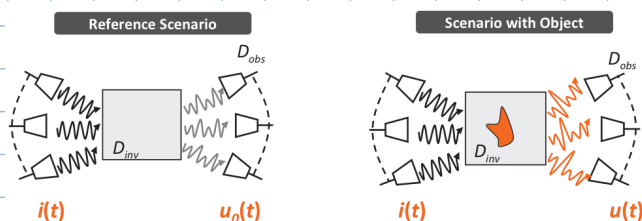


OBJECT

FROM THE EM POV, AN OBJECT (SOMETIMES CALLED OBSTACLE) IS A REGION OF DISCONTINUITY W.R.T. THE REFERENCE SCENARIO.



WE HAVE NOW DEFINED AN OBJECT AS A DISCONTINUITY IN $\epsilon(\underline{r}, t)$, $\mu(\underline{r}, t)$, $\sigma(\underline{r}, t)$, HOWEVER IN ANIRS WE CAN'T DIRECTLY MEASURE THEM, WE HAVE JUST A SIGNAL THAT INTERACTS WITH THEM.



WE CAN THEREFORE SAY THAT A SUFFICIENT CONDITION FOR THE EXISTENCE OF AN OBJECT IS:

$$\Delta \mu(t) := \mu(t) - \mu_0(t) \neq \emptyset$$

MEASURED - REFERENCE

SOURCE

IN THE EM CASE, THE PROBING SOURCE IS MODELLED AS A CURRENT DISTRIBUTION DEFINED IN THE SOURCE DOMAIN $\rightarrow$ ANTENNA.

IT CAN BE DESCRIBED IN TWO WAYS:

- DIRECT DESCRIPTION, FROM THE POV OF THE SOURCE
- INDIRECT DESCRIPTION, FROM THE POV OF THE SCENARIO

DIRECT DESCRIPTION

FROM THE POV OF THE SOURCE, THE SOURCE IS REPRESENTED BY A CURRENT DENSITY FIELD DEFINED IN THE SOURCE DOMAIN

$$\underline{J}(\underline{r}, t) = \begin{cases} \underline{J}_0(\underline{r}, t) & \underline{r} \in D_{\text{source}} \\ 0 & \underline{r} \notin D_{\text{source}} \end{cases} \quad \left[\frac{\text{A}}{\text{m}^2} \right]$$

INDIRECT DESCRIPTION

FROM THE POV OF THE ANIRS SYSTEM HOWEVER, IT IS MORE RELEVANT TO CONSIDER THE SIGNAL THAT IS GENERATED BY THE SOURCE

GENERIC FUNCTION OF SOURCE INVESTIGATION DOMAIN CHARACTERISTICS (VACUUM)

$$\underline{E}_{\text{source}}(\underline{r}, t) = G \left\{ \underline{J}_0(\underline{r}, t); \underline{r} \in D_{\text{source}} \mid (\epsilon_0, \mu_0) \right\}$$

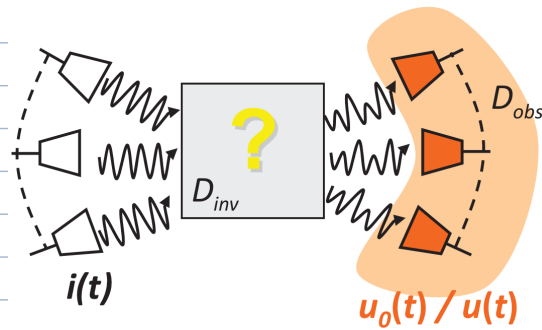
$\underline{r} \in (D_{\text{inv}} \cup D_{\text{obs}}) \rightarrow$ PROPAGATES THROUGH D_{inv} TO D_{obs}

RECALLING THE SYSTEMS MODEL USED IN THE BEGINNING, WE SEE THAT $\underline{E}_{\text{source}}(\underline{r}, t)$ IS THE INPUT.

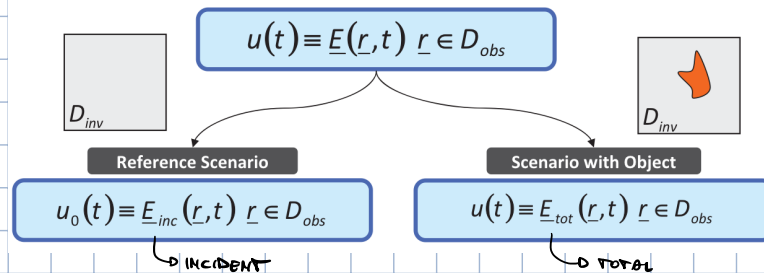
$$i(t) \equiv \underline{E}_{\text{source}}(\underline{r}, t) \quad \forall \underline{r}$$

DETECTOR

IN THE EM CASE, IT IS A SENSOR/ANTENNA CAPABLE OF DETECTING THE EM FIELD BOTH IN PRESENCE AND IN ABSENCE OF THE OBJECT.

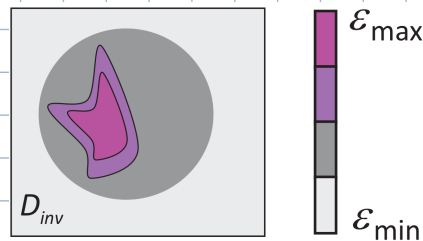


DEPENDING ON WHETHER THE OBJECT IS PRESENT OR NOT, THE DETECTED FIELD TAKES A DIFFERENT NAME:



IMAGING

AN IMAGE IS A COLOR-LEVEL REPRESENTATION OF THE SPATIAL DISTRIBUTION OF A GIVEN QUANTITY. AS A CONSEQUENCE EM IMAGING IS THE PROCESS OF GENERATING A COLOR-LEVEL REPRESENTATION OF THE SPATIAL DISTRIBUTION OF $\epsilon(\underline{r}, t)$, $\mu(\underline{r}, t)$ AND $\sigma(\underline{r}, t)$ IN THE INVESTIGATION DOMAIN.

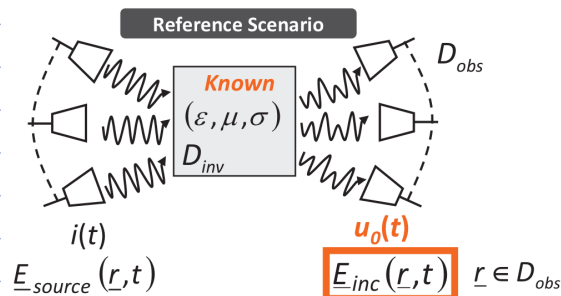


TO DO THIS, WE NEED TO BE ABLE TO DERIVE ϵ, μ, σ FROM $u(t) \rightarrow$ NEED THE RELATION BETWEEN ϵ, μ, σ AND E_{tot} AND SOLVE AN INVERSE PROBLEM.

EM FIELDS

INCIDENT FIELD

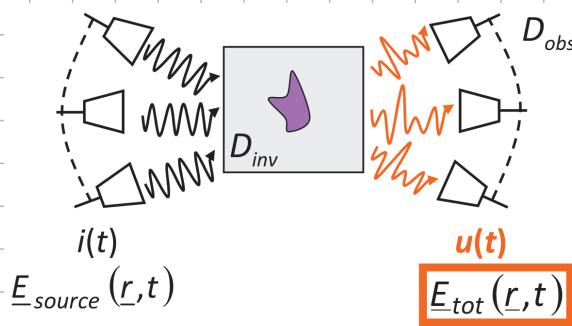
THE INCIDENT FIELD IS DEFINED AS THE EM FIELD IN THE ABSENCE OF AN OBJECT $\Rightarrow$ REFERENCE SCENARIO



$$\begin{aligned} E_{inc}(\underline{r}, t) &= G \left\{ J_0(\underline{r}, t); \underline{r} \in D_{source} \mid (\epsilon, \mu, \sigma) \right\} \\ \underline{r} &\in (D_{inv} \cup D_{obs}) \end{aligned}$$

TOTAL FIELD

THE TOTAL FIELD IS DEFINED AS THE EM FIELD IN THE PRESENCE OF AN OBJECT



$$\underline{E}_{\text{tot}}(\underline{r}, t) = G\left\{\underline{J}_0(\underline{r}, t); \underline{r} \in D_{\text{source}} \left[\underline{\varepsilon}(\underline{r}, t), \underline{\mu}(\underline{r}, t), \underline{\sigma}(\underline{r}, t) \right] \right\}$$

$$\underline{r} \in (D_{\text{inv}} \cup D_{\text{obs}})$$

Now they are not constraints: free space + obs

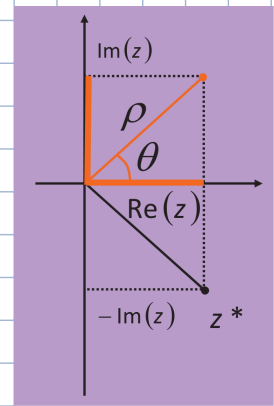
MATHEMATICAL REMINDERS

COMPLEX NUMBERS

$$z = \text{Re}\{z\} + j\text{Im}\{z\} = \rho e^{j\varphi} = \rho [\cos(\varphi) + j\sin(\varphi)]$$

$$\rho = \sqrt{\text{Re}\{z\}^2 + \text{Im}\{z\}^2}$$

$$\varphi = \begin{cases} \tan^{-1} \frac{\text{Im}\{z\}}{\text{Re}\{z\}} & , \text{Re}\{z\} > 0 \\ \tan^{-1} \frac{\text{Im}\{z\}}{\text{Re}\{z\}} + \pi & , \text{Re}\{z\} < 0 \end{cases}$$



PHASORS

IN THE SCALAR CASE, A PHASOR IS A COMPLEX QUANTITY THAT REPRESENTS SINUSOIDAL SIGNALS THAT ABSTRACTS THE TIME DEPENDENCY AND DEPENDS ONLY ON SPATIAL COORDINATES, GIVEN A DETERMINED FREQUENCY.

IN FACT, A SIGNAL CAN BE REPRESENTED BY A PHASOR ONLY IF IT CAN BE EXPRESSED AS SIN/COS.

CONSIDER THE SIGNAL

$$a(t) := A \cos(\omega t + \theta)$$

AND RECALL EULER'S EQUATIONS:

$$e^{j\varphi} = \cos(\varphi) + j\sin(\varphi)$$

$$\cos(\varphi) = \frac{e^{j\varphi} + e^{-j\varphi}}{2}$$

$$\sin(\varphi) = \frac{e^{j\varphi} - e^{-j\varphi}}{2j}$$

WE CAN REPRESENT $a(t)$ AS THE SUM OF TWO COMPLEX FUNCTIONS

$$a(t) = A \cos(\omega t + \theta) = \text{Re}\{A[\cos(\omega t + \theta) + j\sin(\omega t + \theta)]\} = \text{Re}\{Ae^{j(\omega t + \theta)}\} = \text{Re}\{Ae^{j\omega t} e^{j\theta}\} = \text{Re}\{Ae^{j\theta} e^{j\omega t}\} = \text{Re}\{\tilde{A} e^{j\omega t}\}$$

WE DEFINE $\tilde{A}$ AS THE SCALAR PHASOR OF $a(t)$

$$\tilde{A} := Ae^{j\theta}$$

$$a(t) = \text{Re}\{\tilde{A} e^{j\omega t}\}, \quad \tilde{A} := Ae^{j\theta}$$

IN THE VECTORIAL CASE, THE SIGNAL HAS MULTIPLE COMPONENTS.

$$\underline{a}(t) = A_x \cos(\omega t + \theta_x) \hat{x} + A_y \cos(\omega t + \theta_y) \hat{y} + A_z \cos(\omega t + \theta_z) \hat{z}$$

WE CAN REDUCTIVELY ASSUME THE PHASE TO BE EQUAL ON ALL AXES: $\theta_i = \theta \forall i \in \{x, y, z\}$, SO THAT

$$\underline{a}(t) = A_x \cos(\omega t + \theta) \hat{x} + A_y \cos(\omega t + \theta) \hat{y} + A_z \cos(\omega t + \theta) \hat{z}$$

$$= (A_x \hat{x} + A_y \hat{y} + A_z \hat{z}) \cos(\omega t + \theta) = \underline{A} \cos(\omega t + \theta)$$

IN THIS CASE THE PHASORIAL REPRESENTATION HAS A PHASOR FOR EACH AXIS (ESSENTIALLY JUST A DIFFERENT AMPLITUDE $\forall$ AXIS)

$$\underline{a}(t) = \text{Re}\{ \underline{A} e^{j(\omega t + \theta)} \} = \text{Re}\{ \tilde{\underline{A}} e^{j\omega t} \}, \quad \tilde{\underline{A}} = \underline{A} e^{j\theta} = A_p e^{j\theta} \hat{p} \quad \forall p \in \{x, y, z\}$$

UP UNTIL NOW WE HAVE ASSUMED THE AMPLITUDES TO BE CONSTANT IN THE WHOLE DEFINITION SPACE. TO GENERALIZE, WE CAN EXTEND THE DEFINITION TO A VECTOR FIELD.

IN THE CASE OF A TIME-VARYING VECTOR FIELD, WE ADD A SPATIAL DEPENDENCY TO THE AMPLITUDE OF THE SIGNAL

$$\underline{a}(\underline{r}, t) = A_x(\underline{r}) \cos(\omega t + \theta) \hat{x} + A_y(\underline{r}) \cos(\omega t + \theta) \hat{y} + A_z(\underline{r}) \cos(\omega t + \theta) \hat{z}$$

$$= (A_x(\underline{r}) \hat{x} + A_y(\underline{r}) \hat{y} + A_z(\underline{r}) \hat{z}) \cos(\omega t + \theta)$$

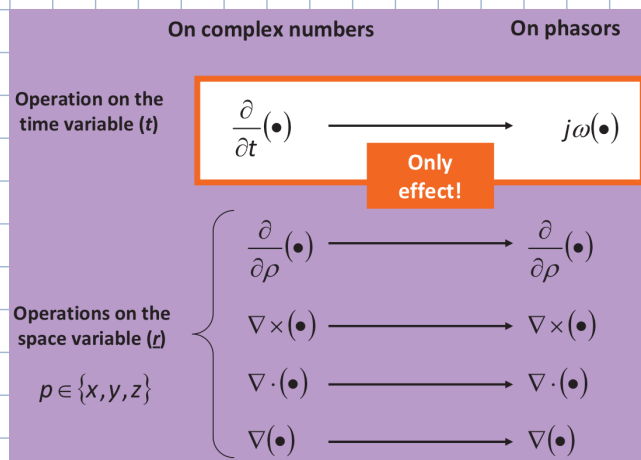
$$= \underline{A}(\underline{r}) \cos(\omega t + \theta)$$

IN THIS CASE THE PHASORIAL REPRESENTATION HAS A SPATIAL DEPENDENCY

$$\underline{a}(\underline{r}, t) = \text{Re}\{ \underline{A}(\underline{r}) e^{j(\omega t + \theta)} \} = \text{Re}\{ \tilde{\underline{A}}(\underline{r}) e^{j\omega t} \}, \quad \tilde{\underline{A}}(\underline{r}) = \underline{A}(\underline{r}) e^{j\theta} = A_p(\underline{r}) e^{j\theta} \hat{p} \quad \forall p \in \{x, y, z\}$$

OPERATIONS WITH PHASORS

THE REASON PHASORS ARE USED IS THAT THEY RENDER TIME-DIFFERENTIAL EQUATIONS LINEAR. ALL OTHER OPERATIONS REMAIN UNTOUCHED.

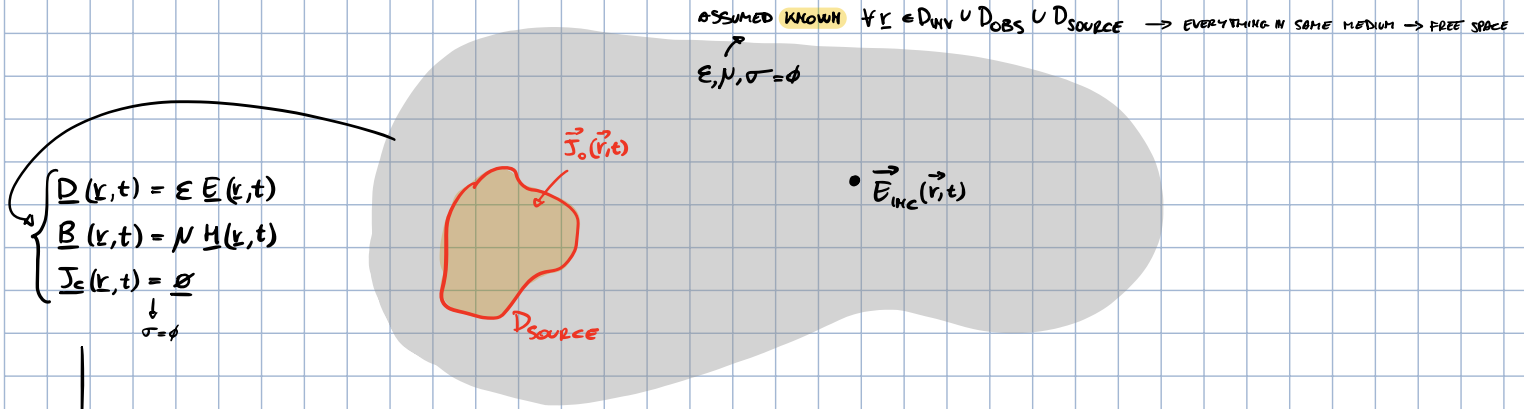


DESCRIPTIVE EQUATIONS

UNTIL NOW WE HAVE WORKED WITH GENERIC SIGNALS $u(t)$, FIELDS $E_{\text{tot}}, E_{\text{inc}}$ AND MEDIUM PARAMETERS. WE NOW NEED TO CLEARLY DEFINE THEIR RELATIONS.

INCIDENT FIELD

BY DEFINITION, THE INCIDENT FIELD IS THE FIELD OF THE REFERENCE SCENARIO (NO OBJECT). OUR GOAL IS TO RELATE THE RADIATED FIELD TO THE SOURCE. WE WANT TO DETERMINE THE FIELD $E_{\text{inc}}(\vec{r}, t)$ RADIATED BY A SOURCE $\vec{J}_0(\vec{r}, t)$ IN FREE-SPACE (NOT NECESSARILY VACUUM, JUST A HOMOGENEOUS REGION WITHOUT OBSTACLES)



THESE OBEY MAXWELL'S EQUATIONS:

$$\begin{cases} \nabla \times E(\vec{r}, t) = -\frac{\partial}{\partial t} B(\vec{r}, t) \\ \nabla \times H(\vec{r}, t) = \frac{\partial}{\partial t} D(\vec{r}, t) + \underline{J}_0(\vec{r}, t) + \underline{J}_c(\vec{r}, t) \end{cases}$$

$\sigma = 0 \Rightarrow \underline{J}_c = 0$

THE GOAL NOW IS TO FIND A SINGLE EQUATION TO REPRESENT THE SOLUTION. WE WANT TO OBTAIN A SOLUTION WITH THE FOLLOWING FORM (PHASORIAL WAVE EQUATION):

$$\begin{cases} \nabla \times \nabla \times \tilde{A}(\vec{r}) - k^2 \tilde{A}(\vec{r}) = c \tilde{B}(\vec{r}) \\ + \text{RADIATION CONDITIONS ON } \tilde{A}(\vec{r}) \end{cases}$$

WHICH HAS THE KNOWN SOLUTION, IF k IS A CONSTANT:

$$\tilde{A}(\vec{r}) = -c \int_{D_{\text{SOURCE}}} \tilde{B}(\vec{r}') \cdot \underline{G}(\vec{r}|\vec{r}') d\vec{r}'$$

$$\nabla \times \underline{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

ELECTRIC FIELD WAVE EQUATION

1) APPLY $\nabla \times \{ \}$ TO $\nabla \times E(\vec{r}, t) = -\frac{\partial}{\partial t} B(\vec{r}, t)$

$$\begin{aligned} \nabla \times \{ \nabla \times E(\vec{r}, t) \} &= \nabla \times \left\{ -\frac{\partial}{\partial t} B(\vec{r}, t) \right\} \\ &\rightarrow \nabla \times \{ \} \text{ HAS NO DEPENDENCY ON TIME. IT IS A SPATIAL-ONLY OPERATOR} \\ \nabla \times \{ \nabla \times E(\vec{r}, t) \} &= -\frac{\partial}{\partial t} [\nabla \times B(\vec{r}, t)] \end{aligned}$$

2) GIVEN $B(\vec{r}, t) = \mu H(\vec{r}, t)$

$$\nabla \times \{ \nabla \times E(\vec{r}, t) \} = -\frac{\partial}{\partial t} [\nabla \times \{ \mu H(\vec{r}, t) \}]$$

$$\nabla \times \{ \nabla \times \underline{E}(\underline{r}, t) \} = -\mu \frac{\partial}{\partial t} [\nabla \times \underline{H}(\underline{r}, t)]$$

3) FROM MAXWELL'S EQUATIONS WE KNOW THAT $\nabla \times \underline{H}(\underline{r}, t) = \frac{\partial}{\partial t} \underline{D}(\underline{r}, t) + \underline{J}_0(\underline{r}, t)$

$$\nabla \times \{ \nabla \times \underline{E}(\underline{r}, t) \} = -\mu \frac{\partial}{\partial t} \left[\frac{\partial}{\partial t} \underline{D}(\underline{r}, t) + \underline{J}_0(\underline{r}, t) \right]$$

4) WE KNOW THAT $\underline{D}(\underline{r}, t) = \epsilon \underline{E}(\underline{r}, t)$

$$\nabla \times \{ \nabla \times \underline{E}(\underline{r}, t) \} = -\mu \frac{\partial}{\partial t} \left[\epsilon \frac{\partial}{\partial t} \underline{E}(\underline{r}, t) + \underline{J}_0(\underline{r}, t) \right]$$

$$\nabla \times \{ \nabla \times \underline{E}(\underline{r}, t) \} = -\mu \epsilon \frac{\partial^2}{\partial t^2} \underline{E}(\underline{r}, t) - \mu \frac{\partial}{\partial t} \underline{J}_0(\underline{r}, t)$$

5) NOW SEPARATE THE UNKNOWN FROM THE KNOWN

DERIVATION IN TIME

$$\nabla \times \{ \nabla \times \underline{E}(\underline{r}, t) \} + \mu \epsilon \frac{\partial^2}{\partial t^2} \underline{E}(\underline{r}, t) = -\mu \frac{\partial}{\partial t} \underline{J}_0(\underline{r}, t)$$

THIS IS THE WAVE EQUATION FOR THE E FIELD. IT'S THE FIELD RADIATED BY THE SOURCE $\underline{J}_0(\underline{r}, t)$ IN FREE SPACE

$$\underline{E}_{\text{INC}}(\underline{r}, t) := \underline{E}(\underline{r}, t)$$

MAGNETIC FIELD WAVE EQUATION

1) APPLY $\nabla \times \{ \cdot \}$ TO $\nabla \times \underline{H}(\underline{r}, t) = \frac{\partial}{\partial t} \underline{D}(\underline{r}, t) + \underline{J}_0(\underline{r}, t)$

$$\nabla \times \{ \nabla \times \underline{H}(\underline{r}, t) \} = \nabla \times \left\{ \frac{\partial}{\partial t} \underline{D}(\underline{r}, t) + \underline{J}_0(\underline{r}, t) \right\}$$

$$\rightarrow \nabla \times \{ \underline{A} + \underline{B} \} = \nabla \times \underline{A} + \nabla \times \underline{B}$$

$$\nabla \times \{ \nabla \times \underline{H}(\underline{r}, t) \} = \frac{\partial}{\partial t} [\nabla \times \underline{D}(\underline{r}, t)] + \nabla \times \underline{J}_0(\underline{r}, t)$$

2) GIVEN $\underline{D}(\underline{r}, t) = \epsilon \underline{E}(\underline{r}, t)$

$$\nabla \times \{ \nabla \times \underline{H}(\underline{r}, t) \} = \epsilon \frac{\partial}{\partial t} [\nabla \times \underline{E}(\underline{r}, t)] + \nabla \times \underline{J}_0(\underline{r}, t)$$

3) FROM MAXWELL'S EQUATIONS WE KNOW THAT $\nabla \times \underline{E}(\underline{r}, t) = -\frac{\partial}{\partial t} \underline{B}(\underline{r}, t)$

$$\nabla \times \{ \nabla \times \underline{H}(\underline{r}, t) \} = \epsilon \frac{\partial}{\partial t} \left[-\frac{\partial}{\partial t} \underline{B}(\underline{r}, t) \right] + \nabla \times \underline{J}_0(\underline{r}, t)$$

$$\nabla \times \{ \nabla \times \underline{H}(\underline{r}, t) \} = -\epsilon \frac{\partial^2}{\partial t^2} \underline{B}(\underline{r}, t) + \nabla \times \underline{J}_0(\underline{r}, t)$$

4) WE KNOW THAT $\underline{B}(\underline{r}, t) = \mu \underline{H}(\underline{r}, t)$

$$\nabla \times \{ \nabla \times \underline{H}(\underline{r}, t) \} = -\mu \epsilon \frac{\partial^2}{\partial t^2} \underline{H}(\underline{r}, t) + \nabla \times \underline{J}_0(\underline{r}, t)$$

5) NOW SEPARATE THE UNKNOWN FROM THE KNOWN

DERIVATION IN SPACE

$$\nabla \times \{ \nabla \times \underline{H}(\underline{r}, t) \} + \mu \epsilon \frac{\partial^2}{\partial t^2} \underline{H}(\underline{r}, t) = \nabla \times \underline{J}_0(\underline{r}, t)$$

THIS IS THE WAVE EQUATION FOR THE H FIELD. IT'S THE FIELD RADIATED BY THE SOURCE $\underline{J}_0(\underline{r}, t)$ IN FREE SPACE

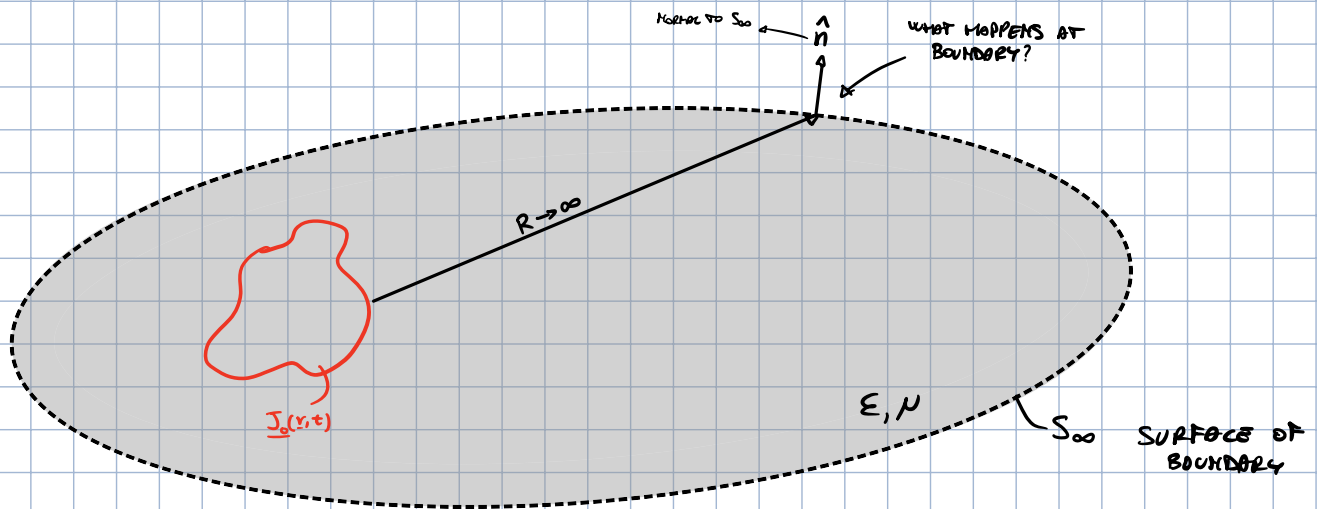
$$\underline{H}_{inc}(\underline{r}, t) := \underline{H}(\underline{r}, t)$$

SOLUTION

TO RECAP, THE EM FIELD $\{E, H, D, B\}$ IN THIS FREE SPACE SCENARIO ($\epsilon, \mu, \sigma = 0$) SATISFIES

$$\begin{cases} \nabla \times \{ \nabla \times \underline{E}_{inc}(\underline{r}, t) \} + \mu \epsilon \frac{\partial^2}{\partial t^2} \underline{E}_{inc}(\underline{r}, t) = -\mu \frac{\partial}{\partial t} \underline{J}_0(\underline{r}, t) \\ \nabla \times \{ \nabla \times \underline{H}_{inc}(\underline{r}, t) \} + \mu \epsilon \frac{\partial^2}{\partial t^2} \underline{H}_{inc}(\underline{r}, t) = \nabla \times \underline{J}_0(\underline{r}, t) \\ \underline{B}_{inc}(\underline{r}, t) = \mu \underline{H}_{inc}(\underline{r}, t) \\ \underline{D}_{inc}(\underline{r}, t) = \epsilon \underline{E}_{inc}(\underline{r}, t) \end{cases}$$

THESE EQUATIONS ARE NOT ENOUGH TO FIND A SOLUTION THOUGH. SINCE THESE ARE DIFFERENTIAL EQUATIONS, WE NEED BOUNDARY CONDITIONS. SINCE WE ARE IN FREE SPACE, THE ONLY BOUNDARY IS AT ∞ .



TO UNDERSTAND WHAT HAPPENS AT THE BOUNDARY, WE USE THE RADIATION CONDITIONS

$$\begin{cases} \underline{E}_{inc}(\underline{r}, t) \times \underline{H}_{inc}(\underline{r}, t) = |\underline{E}_{inc}(\underline{r}, t)| |\underline{H}_{inc}(\underline{r}, t)| \hat{n} = E_{inc}(\underline{r}, t) H_{inc}(\underline{r}, t) \hat{n} \\ \underline{E}_{inc}(\underline{r}, t) = \eta \underline{H}_{inc}(\underline{r}, t) \times \hat{n}, \quad \eta := \sqrt{\frac{\mu}{\epsilon}} \end{cases} \quad \text{valid } \forall \underline{r} \in S_{\infty}$$

E AND H ARE ORTHOGONAL TO EACH OTHER AND TO $\hat{n}$

QUICK REMINDER ON RADIATION CONDITIONS:

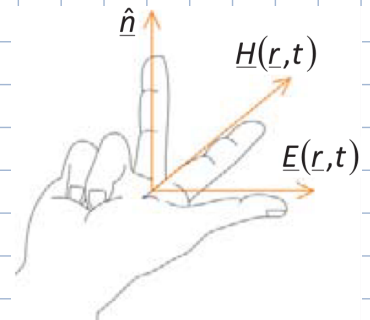
- 1) $\{E, H, \hat{n}\}$ OBEY THE RIGHT HAND RULE (THUMB: E , INDEX: H , MIDDLE: $\hat{n}$)
- 2) EVEN THOUGH AT THE SOURCE THE WAVE IS SPHERICAL, AT S_{∞} IT CAN BE CONSIDERED TO BE A PLANAR WAVE (LOCALLY)
- 3) THE WAVE RADIATED BY THE SOURCE IS PROGRESSIVE ONLY: POWER GOES OUTWARD THROUGH S_{∞}

$$P_{\mu}(t) := \oint_{S_{\infty}} \underline{S}(\underline{r}, t) \cdot \hat{n} ds \geq 0$$

↓
POYNTING VECTOR

4)

$$E(\underline{r}, t) \sim \mathcal{O}\left(\frac{1}{r}\right)$$



$$H(\mathbf{r}, t) \sim O\left(\frac{1}{r}\right)$$

$$S(\mathbf{r}, t) \sim O\left(\frac{1}{r^2}\right)$$

MATHEMATICALLY THE PROBLEM TO BE SOLVED IS THE FOLLOWING

$$\begin{cases} \nabla \times \{ \nabla \times \underline{E}_{inc}(\mathbf{r}, t) \} + \mu \epsilon \frac{\partial^2}{\partial t^2} \underline{E}_{inc}(\mathbf{r}, t) = -\mu \frac{\partial}{\partial t} \underline{J}_0(\mathbf{r}, t) \\ \nabla \times \{ \nabla \times \underline{H}_{inc}(\mathbf{r}, t) \} + \mu \epsilon \frac{\partial^2}{\partial t^2} \underline{H}_{inc}(\mathbf{r}, t) = \nabla \times \underline{J}_0(\mathbf{r}, t) \\ \underline{E}_{inc}(\mathbf{r}, t) \times \underline{H}_{inc}(\mathbf{r}, t) = |\underline{E}_{inc}(\mathbf{r}, t)| |\underline{H}_{inc}(\mathbf{r}, t)| \hat{n} = E_{inc}(\mathbf{r}, t) H_{inc}(\mathbf{r}, t) \hat{n} \\ \underline{E}_{inc}(\mathbf{r}, t) = \eta \underline{H}_{inc}(\mathbf{r}, t) \times \hat{n}, \quad \eta := \sqrt{\frac{\mu}{\epsilon}}, \quad \mathbf{r} \in S_\infty \end{cases}$$

THESE SIGNALS REALLY EXIST, THEREFORE THEY MUST BE **FINITE ENERGY SIGNALS** $\Rightarrow$ **FOURIER TRANSFORM EXISTS.**

THE FOURIER TRANSFORM IS

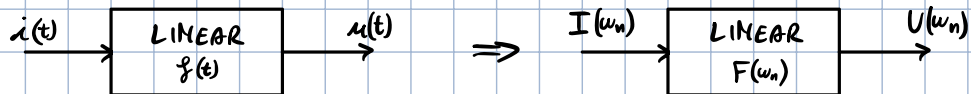
$$\underbrace{F(\omega)}_{\substack{\text{DFT} \\ \text{DFT}}} = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} f(t) e^{-j\omega t} dt \quad \Leftrightarrow \quad f(t) = F^{-1}(\omega) = \int_{\mathbb{R}} F(\omega) e^{j\omega t} d\omega$$

BY ANALYZING THE EXPRESSION AND REMEMBERING THAT $\int \sim \sum$, WE REALIZE THAT A FINITE ENERGY SIGNAL CAN BE EXPRESSED AS A SUM OF **SINUSOIDAL COMPONENTS** $F(\omega_n) e^{j\omega_n t}$.

$$f(t) \approx \frac{1}{\sqrt{2\pi}} \sum_n F(\omega_n) e^{j\omega_n t} \Delta\omega_n$$

TWO THINGS CAN BE NOTED:

- 1) INSTEAD OF ANALYZING EVERYTHING TOGETHER, WE CAN ANALYZE ONE TONE AT A TIME
- 2) BY COMPUTING THE RESPONSE TO A FINITE ENERGY SIGNAL AND KNOWING THAT THE SYSTEM IS LINEAR, WE CAN REWRITE THE SYSTEM IN THE FREQUENCY DOMAIN AND ANALYZE ONE FREQUENCY AT A TIME



IN THIS FRAMEWORK WE START FROM THE INPUT SIGNAL IN TIME, MOVE IN THE FREQUENCY DOMAIN, PASS THROUGH THE LINEAR SYSTEM AND OBTAIN THE OUTPUT SIGNAL IN THE FREQUENCY DOMAIN. SINCE WE ARE STILL INTERESTED IN THE TIME DOMAIN, WE CAN USE THE INVERSE:

$$u(t) = \frac{1}{\sqrt{2\pi}} \sum_n U(\omega_n) e^{j\omega_n t} \Delta\omega_n$$

WE ALSO KNOW THAT, FROM THE CONVOLUTION THEOREM

$$U(\omega_n) = I(\omega_n) F(\omega_n)$$

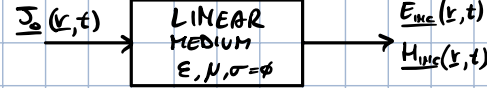
$\hookrightarrow$ IMPULSE RESPONSE OF SYSTEM ASSUMED KNOWN

THEN, WE CAN REWRITE

$$u(t) = \frac{1}{\sqrt{2\pi}} \sum_n I(\omega_n) F(\omega_n) e^{j\omega_n t} \Delta\omega_n$$

LET'S NOW **CONSIDER A SINUSOIDAL TIME DEPENDENCY** FOR ALL SIGNALS.

$$\underline{J}_0(\mathbf{r}, t) := \underline{J}_0(\mathbf{r}) \cos(\omega t + \theta) = \operatorname{Re} \{ \tilde{\underline{J}}_0(\mathbf{r}) e^{j\omega t} \}$$



SINCE THE SYSTEM (MEDIUM) IS LINEAR, THE OUTPUT HAS THE SAME FORM OF THE INPUT. IN FACT WE CAN WRITE THE GENERATED FIELDS AS COSINES:

$$\underline{E}_{inc}(r, t) = \underline{E}_{inc}(r) \cos(\omega t + \theta) = \text{Re}\{\tilde{\underline{E}}_{inc}(r) e^{j\omega t}\}$$

$$\underline{H}_{inc}(r, t) = \underline{H}_{inc}(r) \cos(\omega t + \theta) = \text{Re}\{\tilde{\underline{H}}_{inc}(r) e^{j\omega t}\}$$

NOW, WE CAN FINALLY WRITE THE WAVE EQUATION IN TERMS OF PHASORS, WITH NO DEPENDENCY ON TIME.

$$\begin{cases} \nabla \times \{\nabla \times \underline{E}_{inc}(r, t)\} + \mu \epsilon \frac{\partial^2}{\partial t^2} \underline{E}_{inc}(r, t) = -\mu \frac{\partial}{\partial t} \underline{J}_0(r, t) \\ \nabla \times \{\nabla \times \underline{H}_{inc}(r, t)\} + \mu \epsilon \frac{\partial^2}{\partial t^2} \underline{H}_{inc}(r, t) = \nabla \times \underline{J}_0(r, t) \\ \underline{E}_{inc}(r, t) \times \underline{H}_{inc}(r, t) = |\underline{E}_{inc}(r, t)| |\underline{H}_{inc}(r, t)| \hat{n} = E_{inc}(r, t) H_{inc}(r, t) \hat{n} \\ \underline{E}_{inc}(r, t) = \eta \underline{H}_{inc}(r, t) \times \hat{n}, \quad \eta := \sqrt{\frac{\mu}{\epsilon}}, \quad r \in S_{\infty} \end{cases}$$

$$\begin{cases} \nabla \times \nabla \times \tilde{\underline{E}}_{inc}(r) + \mu \epsilon (j\omega)^2 \tilde{\underline{E}}_{inc}(r) = -\mu j\omega \tilde{\underline{J}}_0(r) \\ \nabla \times \nabla \times \tilde{\underline{H}}_{inc}(r) + \mu \epsilon (j\omega)^2 \tilde{\underline{H}}_{inc}(r) = \nabla \times \tilde{\underline{J}}_0(r) \\ \begin{cases} \tilde{\underline{E}}_{inc}(r) \times \tilde{\underline{H}}_{inc}(r) = |\tilde{\underline{E}}_{inc}(r)| |\tilde{\underline{H}}_{inc}(r)| \hat{n} \\ \tilde{\underline{E}}_{inc}(r) = \eta \tilde{\underline{H}}_{inc}(r) \times \hat{n} \end{cases} \end{cases} = \begin{cases} \nabla \times \nabla \times \tilde{\underline{E}}_{inc}(r) - k^2 \tilde{\underline{E}}_{inc}(r) = -j\omega \mu \tilde{\underline{J}}_0(r) \\ \nabla \times \nabla \times \tilde{\underline{H}}_{inc}(r) - k^2 \tilde{\underline{H}}_{inc}(r) = \nabla \times \tilde{\underline{J}}_0(r) \\ \begin{cases} \tilde{\underline{E}}_{inc}(r) \times \tilde{\underline{H}}_{inc}(r) = |\tilde{\underline{E}}_{inc}(r)| |\tilde{\underline{H}}_{inc}(r)| \hat{n} \\ \tilde{\underline{E}}_{inc}(r) = \eta \tilde{\underline{H}}_{inc}(r) \times \hat{n} \end{cases} \end{cases}$$

• $|\tilde{\underline{H}}_{inc}(r)| = \sqrt{\sum_n \tilde{H}_{inc,n}(r) \tilde{H}_{inc,n}^*(r)}$

BY DEFINING

$$\begin{aligned} k^2 &:= \omega^2 \mu \epsilon, & c &:= -j\omega \mu && \text{FOR THE E FIELD} \\ k^2 &:= \omega^2 \mu \epsilon, & c &:= 1 && \text{FOR THE H FIELD} \end{aligned}$$

$$\hookrightarrow k = \omega \sqrt{\mu \epsilon} \rightarrow \text{RADIATION CONSTANT}$$

WE OBTAIN THE DESIRED FORM OF THE EQUATION

$$\begin{cases} \nabla \times \nabla \times \tilde{\underline{A}}(r) - k^2 \tilde{\underline{A}}(r) = c \tilde{\underline{B}}(r) \\ + \text{RADIATION CONDITIONS ON } \tilde{\underline{A}}(r) \end{cases} \quad \begin{cases} \nabla \times \nabla \times \tilde{\underline{E}}_{inc}(r) - k^2 \tilde{\underline{E}}_{inc}(r) = c \tilde{\underline{J}}_0(r) \\ \nabla \times \nabla \times \tilde{\underline{H}}_{inc}(r) - k^2 \tilde{\underline{H}}_{inc}(r) = c [\nabla \times \tilde{\underline{J}}_0(r)] \\ \begin{cases} \tilde{\underline{E}}_{inc}(r) \times \tilde{\underline{H}}_{inc}(r) = |\tilde{\underline{E}}_{inc}(r)| |\tilde{\underline{H}}_{inc}(r)| \hat{n} \\ \tilde{\underline{E}}_{inc}(r) = \eta \tilde{\underline{H}}_{inc}(r) \times \hat{n} \end{cases} \end{cases}$$

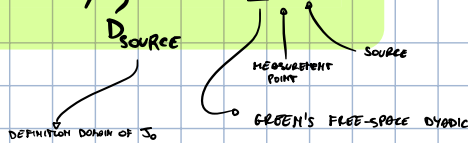
WHICH HAS THE FOLLOWING SOLUTIONS

$$\tilde{\underline{A}}(r) = -c \int_{\text{SOURCE}} \tilde{\underline{B}}(r') \cdot \underline{G}(r|r') dr'$$

∇ ONLY IF k IS A CONSTANT!

$$\tilde{\underline{E}}_{inc}(r) = j\omega \mu \left(\tilde{\underline{J}}_0(r') \cdot \underline{G}(r|r') \right) dr'$$

$$\tilde{\underline{H}}_{inc}(r) = - \left[\nabla \times \tilde{\underline{J}}_0(r') \right] \cdot \underline{G}(r|r') dr'$$



D_{SOURCE}

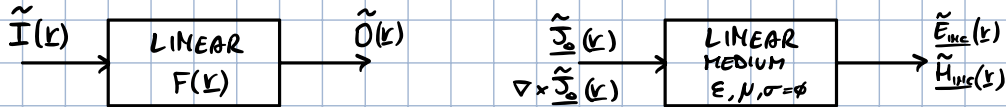
$\underline{G}$ IS FIELD WHEN INPUT CURRENT IS $\underline{J}(r, t)$
 ↳ POINT IN D_{SOURCE} EVALUATE FIELD IN Γ
 AND, USING SUPERPOSITION, SUM UP ALL EFFECTS TO COMPUTE THE ACTUAL FIELD DUE TO THE COMPLETE SOURCE

NOTE THAT THESE RESULTS ARE SIMILAR TO CONVOLUTIONS IF WE CONSIDER $\underline{G}(r, r') = \underline{G}(r - r')$.

GREEN'S FREE-SPACE DYADIC

LET'S NOW TRY TO UNDERSTAND WHAT $\underline{G}(r, r')$ ACTUALLY IS. IN MATHEMATICS, A "GREEN'S FUNCTION" IS THE IMPULSE RESPONSE OF AN INHOMOGENEOUS LINEAR DIFFERENTIAL OPERATOR (~SYSTEM?) DEFINED ON A DOMAIN WITH SPECIFIED BOUNDARY CONDITIONS.

IN OUR CASE IT REPRESENTS THE IMPULSE RESPONSE OF THE MEDIUM (LINEAR → SUPERPOSITION APPLIES)



WE KNOW THAT

$$\tilde{O}(r) = \tilde{I}(r) * \tilde{F}(r)$$

WHICH MEANS WE CAN WRITE

$$\begin{cases} \tilde{E}_{inc}(r) = \tilde{J}_0(r) * \underline{G}(r, r_0) \\ \tilde{H}_{inc}(r) = \nabla \times \tilde{J}_0(r) * \underline{G}(r, r_0) \end{cases}$$

SPATIAL CONVOLUTION

FROM A MATHEMATICAL POV, $\underline{G}$ IS A SECOND ORDER TENSOR (DYADIC).

A VECTOR (FIRST ORDER TENSOR) IS MADE OF A SET OF VALUES RELATED TO A CERTAIN BASIS.

GIVEN A BASIS

$$\{\hat{e}, \hat{m}, \hat{n}\}$$

A VECTOR CAN BE EXPRESSED AS

$$\underline{v} = v_e \hat{e} + v_m \hat{m} + v_n \hat{n} \quad \text{OR} \quad \underline{v} = [v_e, v_m, v_n]^T$$

↳ BASIS IS EXPLICIT

↳ BASIS IS IMPLICIT WITH THIS NOTATION. NO MEANING W/O BASIS

THE DOT PRODUCT OF TWO VECTORS WILL RETURN A SCALAR (THE DOT PRODUCT REDUCES THE ORDER)

$$\underline{v} \cdot \underline{u} = \alpha$$

A DYADIC IS THE RESULT OF THE DYADIC PRODUCT OF TWO VECTORS. THE DYADIC PRODUCT OF TWO VECTORS $\underline{u}, \underline{v}$ IS DENOTED $\underline{u} \underline{v}$ (NO SYMBOLS).

$$\underline{G} = \underline{u} \underline{v} = \sum_j \sum_i u_i v_j \hat{e}_i \hat{e}_j$$

NOT A PRODUCT!

DEFINED GIVEN A BASIS $\{\hat{e}, \hat{m}, \hat{n}\}$

$$\begin{aligned} \underline{u} &= \sum_{i \in \{e, m, n\}} u_i \hat{e}_i \\ \underline{v} &= \sum_{j \in \{e, m, n\}} v_j \hat{e}_j \end{aligned}$$

$$\begin{aligned} &= u_e v_e \hat{e} \hat{e} + u_e v_m \hat{e} \hat{m} + u_e v_n \hat{e} \hat{n} + \\ &\quad u_m v_e \hat{m} \hat{e} + u_m v_m \hat{m} \hat{m} + u_m v_n \hat{m} \hat{n} + \\ &\quad u_n v_e \hat{n} \hat{e} + u_n v_m \hat{n} \hat{m} + u_n v_n \hat{n} \hat{n} \end{aligned}$$

$$= \begin{bmatrix} u_e v_e & u_e v_m & u_e v_n \\ u_m v_e & u_m v_m & u_m v_n \\ u_n v_e & u_n v_m & u_n v_n \end{bmatrix} = \begin{bmatrix} G_{ee} & G_{em} & G_{en} \\ G_{me} & G_{mm} & G_{mn} \\ G_{ne} & G_{nm} & G_{nn} \end{bmatrix}$$

MATRIX REPRESENTATION IS ONLY POSSIBLE IF BASIS IS KNOWN

JUST AS VECTORS HAVE BASIS UNIT VECTORS (VERSORS)

$$\mathbf{i} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{j} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{k} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

THE STANDARD DYADIC BASIS IS REPRESENTED BY

$$\begin{aligned} \mathbf{ii} &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & \mathbf{ij} &= \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & \mathbf{ik} &= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\ \mathbf{ji} &= \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & \mathbf{jj} &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & \mathbf{jk} &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \\ \mathbf{ki} &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, & \mathbf{kj} &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, & \mathbf{kk} &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \end{aligned}$$

$$\underline{\underline{G}} = \underline{\underline{M}} \underline{\underline{N}} = \sum_j \sum_i M_i N_j \hat{\mathbf{i}} \hat{\mathbf{j}} = \sum_j \sum_i G_{ij} \hat{\mathbf{i}} \hat{\mathbf{j}} = \sum_j \underline{\underline{G}}_j \hat{\mathbf{j}}$$

DEFINE AMPHITUDE PRODUCT
 $G_{ij} := M_i N_j$

DEFINE VECTOR ALONG $\hat{\mathbf{j}}$
 $\underline{\underline{G}}_j := \sum_i G_{ij} \hat{\mathbf{i}} = \sum_i \underline{\underline{G}} \cdot \hat{\mathbf{j}}$

PROJECTION OF $\underline{\underline{G}}$ ALONG $\hat{\mathbf{j}}$

THE DOT PRODUCT OF A DYADIC AND A VECTOR RETURNS A VECTOR (DOT PRODUCT REDUCES ORDER OF TENSOR OF HIGHER ORDER)

$$\underline{\underline{G}} \cdot \underline{\underline{V}} = \underline{\underline{M}}$$

FROM AN ELECTROMAGNETIC POV, $\underline{\underline{G}}(\underline{\underline{r}}, \underline{\underline{r}}')$ IS DEFINED AS

$$\underline{\underline{G}}(\underline{\underline{r}}, \underline{\underline{r}}') = -\left(\underline{\underline{I}} + \frac{1}{k^2} \nabla \nabla\right) G(\underline{\underline{r}}, \underline{\underline{r}}')$$

WHERE $G(\underline{\underline{r}}, \underline{\underline{r}}')$ IS GREEN'S FUNCTION FOR FREE SPACE

$$G(\underline{\underline{r}}, \underline{\underline{r}}') = -\frac{1}{4\pi} \frac{e^{-jk|\underline{\underline{r}} - \underline{\underline{r}}'|}}{|\underline{\underline{r}} - \underline{\underline{r}}'|}$$

What is $\underline{\underline{G}}(\underline{\underline{r}})$?
Expand the different components/operators

$$\underline{\underline{G}}(\underline{\underline{r}}) = -\left[\underline{\underline{I}} + \frac{1}{k^2} \nabla \nabla\right] G(\underline{\underline{r}})$$

$\underline{\underline{V}}\underline{\underline{V}} = \underline{\underline{V}}(\underline{\underline{V}})$

$$\underline{\underline{V}} = \begin{pmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \end{pmatrix}$$

Operator applied to...

$$\underline{\underline{V}}\underline{\underline{V}} = \begin{pmatrix} \frac{\partial^2}{\partial x^2} & \frac{\partial^2}{\partial x \partial y} & \frac{\partial^2}{\partial x \partial z} \\ \frac{\partial^2}{\partial y \partial x} & \frac{\partial^2}{\partial y^2} & \frac{\partial^2}{\partial y \partial z} \\ \frac{\partial^2}{\partial z \partial x} & \frac{\partial^2}{\partial z \partial y} & \frac{\partial^2}{\partial z^2} \end{pmatrix}$$

$\underline{\underline{I}} = \underline{\underline{x}}\underline{\underline{x}} + \underline{\underline{y}}\underline{\underline{y}} + \underline{\underline{z}}\underline{\underline{z}}$

$$\underline{\underline{I}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

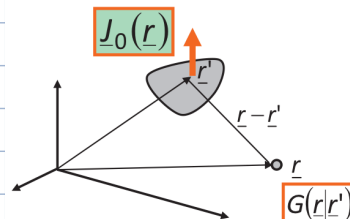
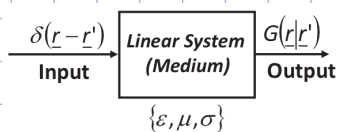
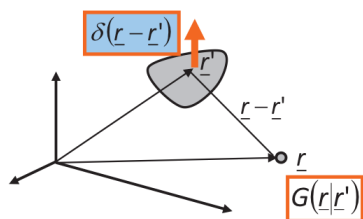
WHICH REPRESENTS A FIELD WHEN THE INPUT CURRENT IS AN IMPULSE: $\underline{\underline{J}}_0(\underline{\underline{r}}, t) = \underline{\underline{J}}(\underline{\underline{r}}, t)$

$$\tilde{f} := e^{-jkz} \rightarrow f(t) = \text{Re}\{\tilde{f} e^{j\omega t}\} = \text{Re}\{e^{-jkz} e^{j\omega t}\} = \text{Re}\{e^{j(\omega t - kz)}\} = \cos(\omega t - kz)$$

WHERE $k := \frac{2\pi}{\lambda}$ IS THE WAVE NUMBER

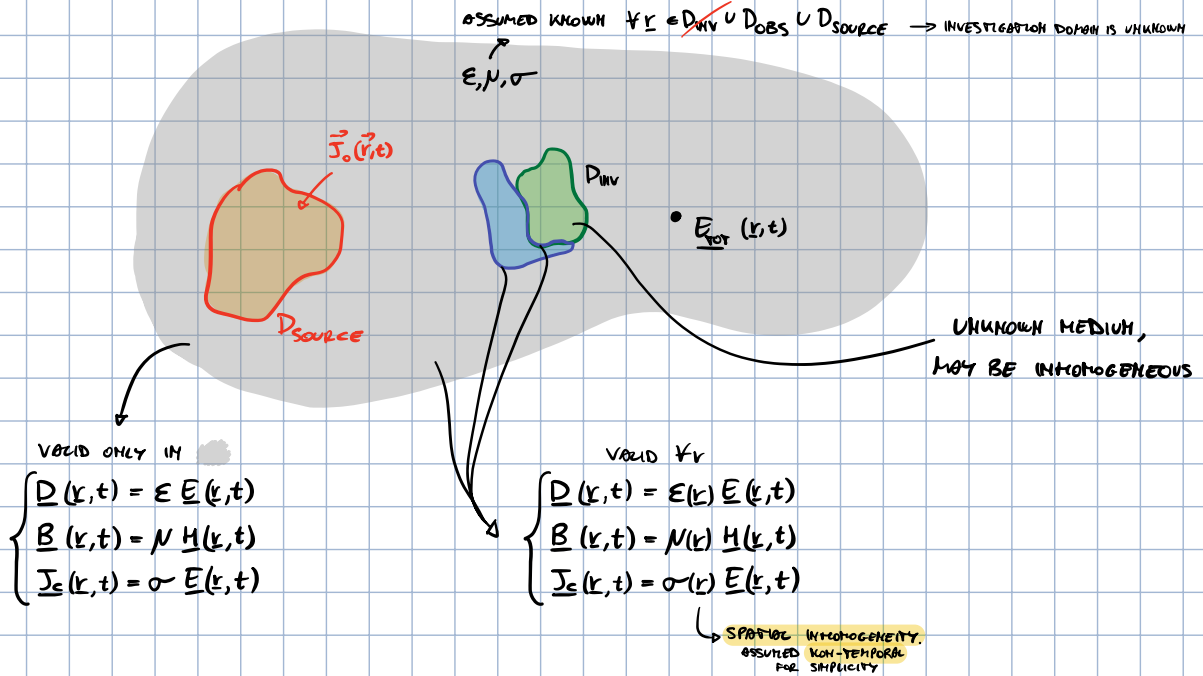
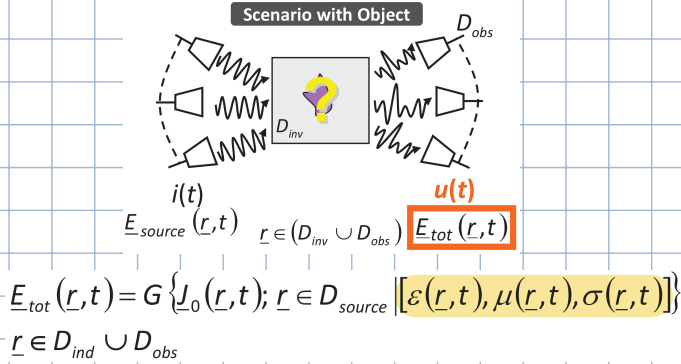
WAVE TRAVELLING ON Z AXIS
 $f(z - vt)$

IN CONCLUSION, $\underline{\underline{G}}(\underline{\underline{r}}, \underline{\underline{r}}')$ IS THE EM FIELD IN $\underline{\underline{r}}$ RADIATED BY AN IMPULSIVE SOURCE $\underline{\underline{J}}_0$ LOCATED IN $\underline{\underline{r}}'$:



TOTAL FIELD

BY DEFINITION, THE TOTAL FIELD IS THE FIELD OF THE SCENARIO WITH THE OBJECT. UNTIL NOW WE STUDIED THE FIELDS IN FREE SPACE, NOW LET'S CONSIDER THE PRESENCE OF AN OBJECT/OBSTACLE (I.E. IN AN INHOMOGENEOUS MEDIUM).
OUR GOAL IS TO DETERMINE THE EM FIELD RADIATED BY A SOURCE $\vec{J}_0(\underline{r}, t)$ IN THE PRESENCE OF AN OBSTACLE $\rightarrow$ DIFFERENT FROM REFERENCE SCENARIO, HAS UNKNOWN FEATURES



IN GENERAL, D_{inv} IS NON-HOMOGENEOUS, THEREFORE THE CONSTITUTIVE EQUATIONS BECOME

$$\underline{D}(\underline{r}, t) = \begin{cases} \epsilon \underline{E}(\underline{r}, t) & , \underline{r} \notin D_{\text{inv}} \\ \epsilon(\underline{r}) \underline{E}(\underline{r}, t) & , \underline{r} \in D_{\text{inv}} \end{cases} \quad \underline{B}(\underline{r}, t) = \begin{cases} \mu \underline{H}(\underline{r}, t) & , \underline{r} \notin D_{\text{inv}} \\ \mu(\underline{r}) \underline{H}(\underline{r}, t) & , \underline{r} \in D_{\text{inv}} \end{cases} \quad \underline{J}_c(\underline{r}, t) = \begin{cases} \sigma \underline{E}(\underline{r}, t) & , \underline{r} \notin D_{\text{inv}} \\ \sigma(\underline{r}) \underline{E}(\underline{r}, t) & , \underline{r} \in D_{\text{inv}} \end{cases}$$

WHICH OBEY MAXWELL'S EQUATIONS

$$\begin{cases} \nabla \times \underline{E}(\underline{r}, t) = - \frac{\partial}{\partial t} \underline{B}(\underline{r}, t) \\ \nabla \times \underline{H}(\underline{r}, t) = \frac{\partial}{\partial t} \underline{D}(\underline{r}, t) + \underline{J}_0(\underline{r}, t) + \underline{J}_c(\underline{r}, t) \end{cases}$$

NEW! $\Leftarrow \sigma \neq 0$

AS WITH THE INCIDENT FIELD, THE GOAL NOW IS TO FIND A SINGLE EQUATION TO REPRESENT THE SOLUTION. WE WANT TO OBTAIN A SOLUTION WITH THE FOLLOWING FORM (PHASORIAL WAVE EQUATION):

$$\begin{cases} \nabla \times \nabla \times \tilde{\underline{A}}(\underline{r}) - k^2 \tilde{\underline{A}}(\underline{r}) = \underline{c} \tilde{\underline{B}}(\underline{r}) \\ + \text{RADIATION CONDITIONS ON } \tilde{\underline{A}}(\underline{r}) \end{cases}$$

WHICH HAS THE KNOWN SOLUTION, IF k IS A CONSTANT:

$$\tilde{A}(\underline{r}) = -c \int_{D_{\text{source}}} \tilde{B}(\underline{r}') \cdot \underline{G}(\underline{r}|\underline{r}') d\underline{r}'$$

ELECTRIC FIELD WAVE EQUATION

1) APPLY $\nabla \times \{ \cdot \}$ TO $\nabla \times \underline{E}(\underline{r}, t) = -\frac{\partial}{\partial t} \underline{B}(\underline{r}, t)$

$$\nabla \times \{ \nabla \times \underline{E}(\underline{r}, t) \} = \nabla \times \left\{ -\frac{\partial}{\partial t} \underline{B}(\underline{r}, t) \right\}$$

$\nabla \times \{ \cdot \}$ HAS NO DEPENDENCY ON TIME.
IT IS A SPATIAL-ONLY OPERATOR

$$\nabla \times \{ \nabla \times \underline{E}(\underline{r}, t) \} = -\frac{\partial}{\partial t} [\nabla \times \underline{B}(\underline{r}, t)]$$

2) GIVEN $\underline{B}(\underline{r}, t) = \mu(\underline{r}) \underline{H}(\underline{r}, t)$

$$\nabla \times \{ \nabla \times \underline{E}(\underline{r}, t) \} = -\frac{\partial}{\partial t} [\nabla \times \{ \mu(\underline{r}) \underline{H}(\underline{r}, t) \}]$$

$$\nabla \times \{ \nabla \times \underline{E}(\underline{r}, t) \} = -\mu(\underline{r}) \frac{\partial}{\partial t} [\nabla \times \underline{H}(\underline{r}, t)]$$

3) FROM MAXWELL'S EQUATIONS WE KNOW THAT $\nabla \times \underline{H}(\underline{r}, t) = \frac{\partial}{\partial t} \underline{D}(\underline{r}, t) + \underline{J}_0(\underline{r}, t) + \underline{J}_c(\underline{r}, t)$

$$\nabla \times \{ \nabla \times \underline{E}(\underline{r}, t) \} = -\mu(\underline{r}) \frac{\partial}{\partial t} \left[\frac{\partial}{\partial t} \underline{D}(\underline{r}, t) + \underline{J}_0(\underline{r}, t) + \underline{J}_c(\underline{r}, t) \right]$$

4) WE KNOW THAT $\underline{D}(\underline{r}, t) = \epsilon(\underline{r}) \underline{E}(\underline{r}, t)$ AND $\underline{J}_c(\underline{r}, t) = \sigma(\underline{r}) \underline{E}(\underline{r}, t)$

$$\nabla \times \{ \nabla \times \underline{E}(\underline{r}, t) \} = -\mu(\underline{r}) \frac{\partial}{\partial t} \left[\frac{\partial}{\partial t} \epsilon(\underline{r}) \underline{E}(\underline{r}, t) + \underline{J}_0(\underline{r}, t) + \sigma(\underline{r}) \underline{E}(\underline{r}, t) \right]$$

$$\nabla \times \{ \nabla \times \underline{E}(\underline{r}, t) \} = -\mu(\underline{r}) \epsilon(\underline{r}) \frac{\partial^2}{\partial t^2} \underline{E}(\underline{r}, t) - \mu(\underline{r}) \frac{\partial}{\partial t} \underline{J}_0(\underline{r}, t) - \mu(\underline{r}) \sigma(\underline{r}) \frac{\partial}{\partial t} \underline{E}(\underline{r}, t)$$

5) NOW SEPARATE THE UNKNOWN FROM THE KNOWN

$$\nabla \times \{ \nabla \times \underline{E}(\underline{r}, t) \} + \mu(\underline{r}) \epsilon(\underline{r}) \frac{\partial^2}{\partial t^2} \underline{E}(\underline{r}, t) + \mu(\underline{r}) \sigma(\underline{r}) \frac{\partial}{\partial t} \underline{E}(\underline{r}, t) = -\mu(\underline{r}) \frac{\partial}{\partial t} \underline{J}_0(\underline{r}, t)$$

THIS IS THE WAVE EQUATION FOR THE $\underline{E}$ FIELD. IT'S THE FIELD RADIATED BY THE SOURCE $\underline{J}_0(\underline{r}, t)$ IN FREE SPACE

$$\underline{E}_{\text{tot}}(\underline{r}, t) := \underline{E}(\underline{r}, t)$$

MAGNETIC FIELD WAVE EQUATION

1) APPLY $\nabla \times \{ \cdot \}$ TO $\nabla \times \underline{H}(\underline{r}, t) = \frac{\partial}{\partial t} \underline{D}(\underline{r}, t) + \underline{J}_0(\underline{r}, t) + \underline{J}_c(\underline{r}, t)$

$$\nabla \times \{ \nabla \times \underline{H}(\underline{r}, t) \} = \nabla \times \left\{ \frac{\partial}{\partial t} \underline{D}(\underline{r}, t) + \underline{J}_0(\underline{r}, t) + \underline{J}_c(\underline{r}, t) \right\}$$

$\nabla \times \{ \underline{A} + \underline{B} \} = \nabla \times \underline{A} + \nabla \times \underline{B}$

$$\nabla \times \{ \nabla \times \underline{H}(\underline{r}, t) \} = \frac{\partial}{\partial t} [\nabla \times \underline{D}(\underline{r}, t)] + \nabla \times \underline{J}_0(\underline{r}, t) + \nabla \times \underline{J}_c(\underline{r}, t)$$

2) GIVEN $\underline{D}(\underline{r}, t) = \epsilon(\underline{r}) \underline{E}(\underline{r}, t)$ AND $\underline{J}_c(\underline{r}, t) = \sigma(\underline{r}) \underline{E}(\underline{r}, t)$

$$\nabla \times \{ \nabla \times \underline{H}(\underline{r}, t) \} = \frac{\partial}{\partial t} [\nabla \times \epsilon(\underline{r}) \underline{E}(\underline{r}, t)] + \nabla \times \underline{J}_0(\underline{r}, t) + \nabla \times [\sigma(\underline{r}) \underline{E}(\underline{r}, t)]$$

$$\nabla \times \{ \nabla \times \underline{H}(\underline{r}, t) \} = \epsilon(\underline{r}) \frac{\partial^2}{\partial t^2} [\nabla \times \underline{E}(\underline{r}, t)] + \nabla \times \underline{J}_0(\underline{r}, t) + \sigma(\underline{r}) \nabla \times \underline{E}(\underline{r}, t)$$

3) FROM MAXWELL'S EQUATIONS WE KNOW THAT $\nabla \times \underline{E}(\underline{r}, t) = -\frac{\partial}{\partial t} \underline{B}(\underline{r}, t)$

$$\nabla \times \{ \nabla \times \underline{H}(\underline{r}, t) \} = \epsilon(\underline{r}) \frac{\partial^2}{\partial t^2} \left[-\frac{\partial}{\partial t} \underline{B}(\underline{r}, t) \right] + \nabla \times \underline{J}_0(\underline{r}, t) + \sigma(\underline{r}) \nabla \times \left[-\frac{\partial}{\partial t} \underline{B}(\underline{r}, t) \right]$$

$$\nabla \times \{ \nabla \times \underline{H}(\underline{r}, t) \} = -\epsilon(\underline{r}) \frac{\partial^2}{\partial t^2} \underline{B}(\underline{r}, t) + \nabla \times \underline{J}_0(\underline{r}, t) - \sigma(\underline{r}) \frac{\partial}{\partial t} \nabla \times \underline{B}(\underline{r}, t)$$

4) WE KNOW THAT $\underline{B}(\underline{r}, t) = \mu(\underline{r}) \underline{H}(\underline{r}, t)$

$$\nabla \times \{ \nabla \times \underline{H}(\underline{r}, t) \} = -\epsilon(\underline{r}) \frac{\partial^2}{\partial t^2} \mu(\underline{r}) \underline{H}(\underline{r}, t) + \nabla \times \underline{J}_0(\underline{r}, t) - \sigma(\underline{r}) \frac{\partial}{\partial t} \nabla \times [\mu(\underline{r}) \underline{H}(\underline{r}, t)]$$

$$\nabla \times \{ \nabla \times \underline{H}(\underline{r}, t) \} = -\mu(\underline{r}) \epsilon(\underline{r}) \frac{\partial^2}{\partial t^2} \underline{H}(\underline{r}, t) + \nabla \times \underline{J}_0(\underline{r}, t) - \mu(\underline{r}) \sigma(\underline{r}) \frac{\partial}{\partial t} \nabla \times \underline{H}(\underline{r}, t)$$

5) NOW SEPARATE THE UNKNOWN FROM THE KNOWN

$$\nabla \times \{ \nabla \times \underline{H}(\underline{r}, t) \} + \mu(\underline{r}) \epsilon(\underline{r}) \frac{\partial^2}{\partial t^2} \underline{H}(\underline{r}, t) + \mu(\underline{r}) \sigma(\underline{r}) \frac{\partial}{\partial t} \nabla \times \underline{H}(\underline{r}, t) = \nabla \times \underline{J}_0(\underline{r}, t)$$

THIS IS THE WAVE EQUATION FOR THE $\underline{H}$ FIELD. IT'S THE FIELD RADIATED BY THE SOURCE $\underline{J}_0(\underline{r}, t)$ IN FREE SPACE

$$\underline{H}_{\text{tot}}(\underline{r}, t) := \underline{H}(\underline{r}, t)$$

SOLUTION

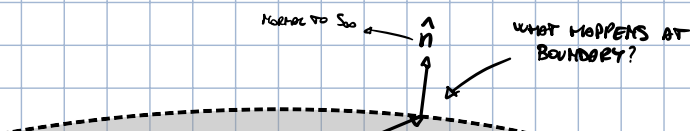
TO RECAP, THE EM FIELD $\{E, H, D, B\}$ IN THIS SCENARIO WITH OBJECT (ϵ, μ, σ) SATISFIES

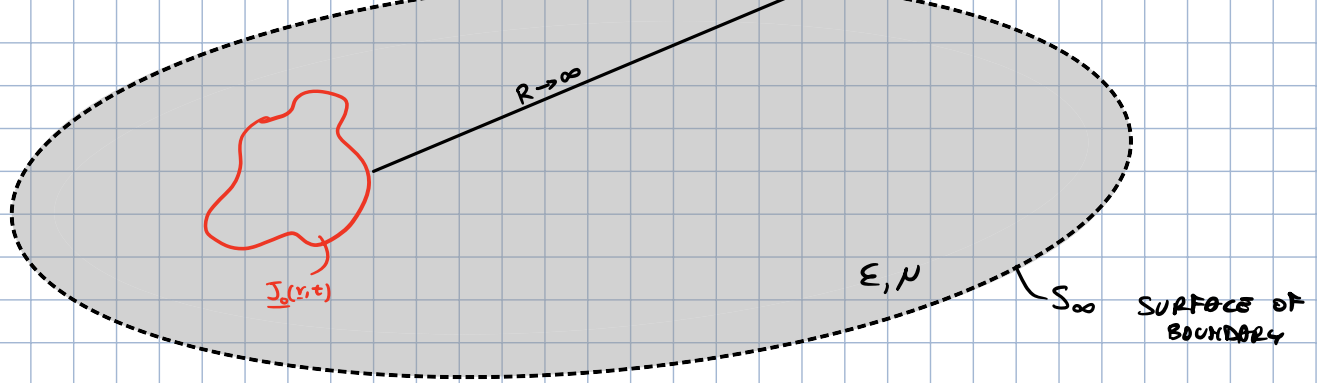
NOTE: SIMPLIFY $\mu(\underline{r}) = \mu$, $\sigma(\underline{r}) = \sigma$ ($\epsilon(\underline{r})$ STILL DEPENDS ON $\underline{r}$)

$$\left\{ \begin{array}{l} \nabla \times \nabla \times \underline{E}_{\text{tot}}(\underline{r}, t) + \mu \epsilon(\underline{r}) \frac{\partial^2}{\partial t^2} \underline{E}_{\text{tot}}(\underline{r}, t) + \mu \sigma \frac{\partial}{\partial t} \underline{E}_{\text{tot}}(\underline{r}, t) = -\mu \frac{\partial}{\partial t} \underline{J}_0(\underline{r}, t) \\ \nabla \times \nabla \times \underline{H}_{\text{tot}}(\underline{r}, t) + \mu \epsilon(\underline{r}) \frac{\partial^2}{\partial t^2} \underline{H}_{\text{tot}}(\underline{r}, t) + \mu \sigma \frac{\partial}{\partial t} \nabla \times \underline{H}_{\text{tot}}(\underline{r}, t) = \nabla \times \underline{J}_0(\underline{r}, t) \\ \underline{B}_{\text{tot}}(\underline{r}, t) = \mu \underline{H}_{\text{tot}}(\underline{r}, t) \\ \underline{D}_{\text{tot}}(\underline{r}, t) = \epsilon(\underline{r}) \underline{E}_{\text{tot}}(\underline{r}, t) \\ \underline{J}_c(\underline{r}, t) = \sigma \underline{E}_{\text{tot}}(\underline{r}, t) \end{array} \right.$$

THESE EQUATIONS ARE NOT ENOUGH TO FIND A SOLUTION THOUGH. SINCE THESE ARE DIFFERENTIAL EQUATIONS, WE NEED BOUNDARY CONDITIONS. NOW WE HAVE MULTIPLE BOUNDARIES! $\rightarrow$ DIFFERENT FROM FREE-SPACE

1) BOUNDARY CONDITION AT ∞ (SAME AS FREE-SPACE)



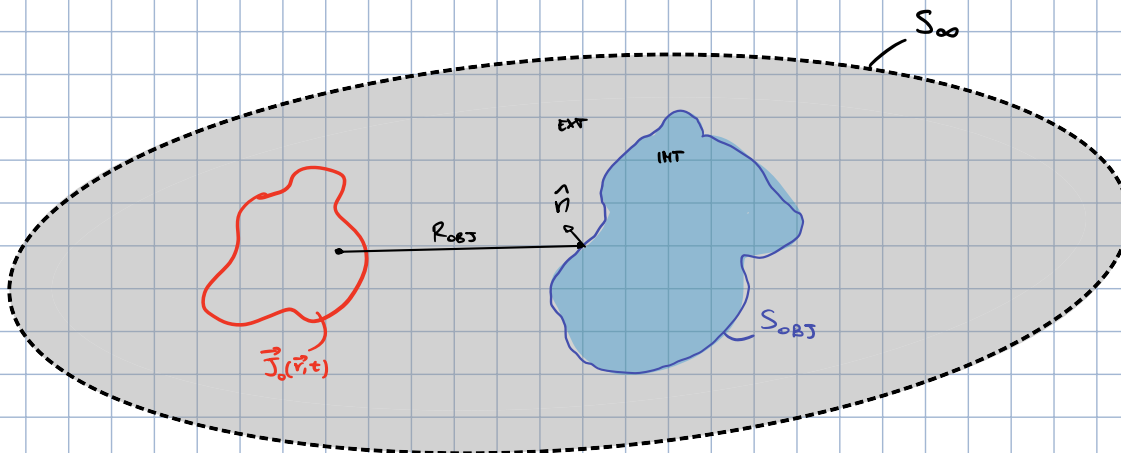


TO UNDERSTAND WHAT HAPPENS AT THE BOUNDARY, WE USE THE RADIATION CONDITIONS

$$\begin{cases} \underline{E}_{\text{INC}}(\vec{r}, t) \times \underline{H}_{\text{INC}}(\vec{r}, t) = |\underline{E}_{\text{INC}}(\vec{r}, t)| |\underline{H}_{\text{INC}}(\vec{r}, t)| \hat{n} = E_{\text{INC}}(\vec{r}, t) H_{\text{INC}}(\vec{r}, t) \hat{n} \\ \underline{E}_{\text{INC}}(\vec{r}, t) = \eta \underline{H}_{\text{INC}}(\vec{r}, t) \times \hat{n}, \quad \eta := \sqrt{\frac{\mu}{\epsilon}} \end{cases} \quad \text{valid } \forall \vec{r} \in S_\infty$$

E AND H ARE ORTHOGONAL TO EACH OTHER AND TO $\hat{n}$

2) BOUNDARY CONDITIONS AT OBJECT (ASSUMED ONLY ONE BOUNDARY FOR SIMPLICITY)



$$\begin{cases} \hat{n} \times \underline{E}_{\text{TOT}}^{\text{EXT}}(\vec{r}, t) = \hat{n} \times \underline{E}_{\text{TOT}}^{\text{INT}}(\vec{r}, t) \\ \hat{n} \times \underline{H}_{\text{TOT}}^{\text{EXT}}(\vec{r}, t) - \hat{n} \times \underline{H}_{\text{TOT}}^{\text{INT}}(\vec{r}, t) = \underline{J}_s(\vec{r}, t) \\ \hat{n} \cdot \underline{D}_{\text{TOT}}^{\text{EXT}}(\vec{r}, t) - \hat{n} \cdot \underline{D}_{\text{TOT}}^{\text{INT}}(\vec{r}, t) = \rho_s(\vec{r}, t) \\ \hat{n} \cdot \underline{B}_{\text{TOT}}^{\text{EXT}}(\vec{r}, t) = \hat{n} \cdot \underline{B}_{\text{TOT}}^{\text{INT}}(\vec{r}, t) \end{cases}$$

TANGENTIAL COMPONENT OF E IS THE SAME OUTSIDE AND INSIDE

TANGENTIAL COMPONENT OF H DIFFERS BETWEEN OUTSIDE AND INSIDE BY AN AMOUNT GIVEN BY $\underline{J}_s \rightarrow$ DISCONTINUITY INDUCES SURFACE CURRENT

NORMAL COMPONENT OF D DIFFERS BETWEEN OUTSIDE AND INSIDE BY AN AMOUNT GIVEN BY ρ_s

NORMAL COMPONENT OF B IS THE SAME OUTSIDE AND INSIDE

LET'S NOW CONSIDER A SINUSOIDAL TIME DEPENDENCY FOR ALL SIGNALS.

$$\underline{J}_s(\vec{r}, t) := \underline{J}_s(\vec{r}) \cos(\omega t + \theta) = \text{Re} \{ \tilde{\underline{J}}_s(\vec{r}) e^{j\omega t} \}$$

$$\underline{E}_{\text{TOT}}(\vec{r}, t) = \underline{E}_{\text{INC}}(\vec{r}) \cos(\omega t + \theta) = \text{Re} \{ \tilde{\underline{E}}_{\text{TOT}}(\vec{r}) e^{j\omega t} \}$$

$$\underline{H}_{\text{TOT}}(\vec{r}, t) = \underline{H}_{\text{INC}}(\vec{r}) \cos(\omega t + \theta) = \text{Re} \{ \tilde{\underline{H}}_{\text{TOT}}(\vec{r}) e^{j\omega t} \}$$

WE CAN THEN REWRITE THE EQUATIONS USING PHASORS

$$\nabla \times \nabla \times \underline{E}_{\text{tot}}(\underline{r}, t) + \mu \epsilon(\underline{r}) \frac{\partial^2}{\partial t^2} \underline{E}_{\text{tot}}(\underline{r}, t) + \mu \sigma \frac{\partial}{\partial t} \underline{E}_{\text{tot}}(\underline{r}, t) = -\mu \frac{\partial^2}{\partial t^2} \underline{J}_0(\underline{r}, t)$$

$$\nabla \times \nabla \times \underline{H}_{\text{tot}}(\underline{r}, t) + \mu \epsilon(\underline{r}) \frac{\partial^2}{\partial t^2} \underline{H}_{\text{tot}}(\underline{r}, t) + \mu \sigma \frac{\partial}{\partial t} \nabla \times \underline{H}_{\text{tot}}(\underline{r}, t) = \nabla \times \underline{J}_0(\underline{r}, t)$$

↓

$$\begin{cases} \nabla \times \nabla \times \tilde{\underline{E}}_{\text{tot}}(\underline{r}) - \omega^2 \mu \epsilon(\underline{r}) \tilde{\underline{E}}_{\text{tot}}(\underline{r}) + j\omega \mu \sigma \tilde{\underline{E}}_{\text{tot}}(\underline{r}) = -j\omega \mu \tilde{\underline{J}}_0(\underline{r}) \\ \nabla \times \nabla \times \tilde{\underline{H}}_{\text{tot}}(\underline{r}) - \omega^2 \mu \epsilon(\underline{r}) \tilde{\underline{H}}_{\text{tot}}(\underline{r}) + j\omega \mu \sigma \nabla \times \tilde{\underline{H}}_{\text{tot}}(\underline{r}) = \nabla \times \tilde{\underline{J}}_0(\underline{r}) \end{cases}$$

BY GROUPING $\underline{E}_{\text{tot}}$:

$$\nabla \times \nabla \times \tilde{\underline{E}}_{\text{tot}}(\underline{r}) - \omega^2 \mu \left[\epsilon(\underline{r}) + j \frac{\sigma}{\omega} \right] \tilde{\underline{E}}_{\text{tot}}(\underline{r}) = -j\omega \mu \tilde{\underline{J}}_0(\underline{r})$$

FROM WHICH WE DEFINE THE COMPLEX EPSILON

$$\tilde{\epsilon}(\underline{r}) := \epsilon(\underline{r}) + j \frac{\sigma}{\omega}$$

REMEMBERING THE FORM WE WANT TO OBTAIN

$$\begin{cases} \nabla \times \nabla \times \tilde{\underline{A}}(\underline{r}) - k^2 \tilde{\underline{A}}(\underline{r}) = c \tilde{\underline{B}}(\underline{r}) \\ + \text{RADIATION CONDITIONS ON } \tilde{\underline{A}}(\underline{r}) \end{cases}$$

WE CAN DEFINE THE WAVE NUMBER k

$$\nabla \times \nabla \times \tilde{\underline{E}}_{\text{tot}}(\underline{r}) - \omega^2 \mu \left[\epsilon(\underline{r}) + j \frac{\sigma}{\omega} \right] \tilde{\underline{E}}_{\text{tot}}(\underline{r}) = -j\omega \mu \tilde{\underline{J}}_0(\underline{r})$$

$$k^2(\underline{r}) := \omega^2 \mu \left[\epsilon(\underline{r}) + j \frac{\sigma}{\omega} \right] = \omega^2 \mu \tilde{\epsilon}(\underline{r})$$

THIS IS NO LONGER A CONSTANT! CAN'T USE SAME SOLUTION OF INCIDENT FIELD

$$\tilde{\underline{A}}(\underline{r}) = -c \int_{D_{\text{SOURCE}}} \tilde{\underline{B}}(\underline{r}') \cdot \underline{G}(\underline{r}|\underline{r}') d\underline{r}'$$

SO HOW DO WE DETERMINE THE TOTAL FIELD?

↳ EQUIVALENT PROBLEM